

PhytoGraph — cycles 47-49

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PhytoGraph — cycles 47-49

Abstract

Cycles 47-49 preserved PhytoGraph’s terminal maintenance state. The supplied audit report records no critical findings, no moderate findings, and no new actionable minor findings. It validates the worker posture as a no-op: no concrete defect, reproducibility failure, public-facing issue, or qualifying scientific evidence was supplied, so no artifact rebuild, file edit, or ledger promotion was warranted during worker/auditor cycle work.

The audit decision was `VALIDATED`. The master `prediction_ledger.tsv` and `speculation_ledger.tsv` each remained header-only with 1 line. `promise_ledger.jsonl` contained 239 events. `promise_check` exited 0 with 239 events and 46 plan milestones, and `org_check` exited 0 with inherited root-layout warnings only.

No track reopening predicate was met. All six tracks remain closed under the previously audited limitation statuses. The original campaign success criterion of at least one validated prediction per track remains unmet, and PhytoGraph remains closed as a conservative infrastructure, validation, falsification, public communication, and handoff package.

Introduction

PhytoGraph’s terminal state is governed by the master-ledger promotion rule. A track-local result, Atlas display, closure note, or maintenance artifact does not become a campaign-level prediction unless it is promoted into `prediction_ledger.tsv` under the validation protocol. A claim that cannot name a validation source belongs in `speculation_ledger.tsv`. Neither master ledger received data rows during cycles 47-49.

The Live Recovery Addendum directed continuation from the last substantive state represented by `reports/cycles/report_cycles_17-19.md` and the post-merge integration artifacts for fork `cc044bf40be3`, while treating later no-work continuity reports as non-scientific progress. Cycles 44-46 already preserved terminal closure under a validated no-op maintenance posture. Cycles 47-49 continue that same posture.

This report consolidates cycles 47-49. It reports the supplied audit and local project state. It does not re-audit the scientific results, reopen tracks, or re-derive validation outcomes.

Approach

The primary source for this report is the supplied cycles 47-49 audit report. Local workspace artifacts were used for continuity and simple sanity checks: MANIFEST.md, prediction_ledger.tsv, speculation_ledger.tsv, promise_ledger.jsonl, REFERENCES.md, reports/cycles/report_cycles_44-46.md, reports/final_campaign_handoff_manifest.md, reports/taxonomy_results_site_closure_note.md, and reports/taxonomy_results_site_qa.md.

The supplied cycle-session IDs were:

Cycle	Researcher	Worker	Auditor
47	a28c3d4e-df0b-4dde-9b0f-3c97a65ff65f	e5ee4ff7-5955-4b6b-aad2-9813acc535b8	3bef1ab3-8c79-4791-9264-68605fd96636
48	f8e1196a-8ac8-485f-a523-6e27cb4d44fd	2be5d2a6-93dc-4cd4-a08e-3ba807069137	6a024481-6a54-4045-b469-efbe3e3746ae
49	e34330da-c6c0-49fb-8844-9ae9af65eedf	865c46d2-6168-48e5-b96e-4d05b5c377d3	b0ec19aa-195d-4f75-bf11-444d8558bdac

Full native transcripts for these sessions could not be fetched in this environment because session-search/session-fetch tools were not available. That is a record gap. The report therefore relies on the supplied audit and local workspace artifacts.

No report markdown was written to disk by the reporter. Per protocol, this report content is supplied in the output block for the harness to write and render.

Findings

Validated No-Op Maintenance Decision

The central finding for cycles 47-49 is a validated no-op. The audit states that the cycle inputs provided no concrete maintenance defect, reproducibility failure, public-facing clarity/accessibility or broken-link issue, or qualifying scientific evidence package. Because no trigger predicate was present, the worker correctly avoided reopening tracks, rebuilding artifacts, editing files, or promoting ledger rows.

This is a maintenance finding, not a new scientific result. It adds no prediction, validation, falsification, benchmark, schema change, Atlas change, or track-local output.

Track Reopening Eligibility

No track reopening predicate was met. The six final track outcomes remain preserved:

Track	Preserved status	Meaning
Track 1 Reticulation Atlas	sidecar_readiness_uncontrolled	Reticulation-sidecar readiness remains insufficient for master promotion because accepted-key recovery and control requirements remain unresolved.
Track 2 Ghost Hyperedges	H2_remains_not_supported_or_data_limited	Held-out anachronism recovery remains unsupported or data-limited under the campaign validation rule.
Track 3 Convergence Pressure	confound_limited	Convergence-pressure evidence remains limited by family-size and sampling-density confounds.
Track 4 Domestication Hypergraph	still_data_limited	Crop-wild-relative and climate-envelope evidence remains insufficient for recommendation-like substitution claims.
Track 5 Chemodiversity Predictor	H5_remains_source_biased	Temporal phytochemistry validation remains source-biased and insufficient for promotion.
Track 6 Foundation Model Probe	environment_limited_untested	The static benchmark and deterministic harness remain deliverables, but no qualifying local/open model-response evaluation was promoted.

These statuses are preserved from prior closure work. Cycles 47-49 supplied no new evidence that would change any track status.

Master Ledgers

The master ledgers remained non-promoted:

Ledger	Count	Status
prediction_ledger.tsv	1 line	Header-only
speculation_ledger.tsv	1 line	Header-only
promise_ledger.jsonl	239 lines	Append-only event ledger

The reporter read the master prediction and speculation ledgers and confirmed that each file contains only its header. This matches the supplied audit statement that no master prediction or speculation rows were promoted.

Validators And Warnings

The supplied audit reports these checks:

Check	Result
<code>wc -l prediction_ledger.tsv</code>	1, 1, and 239
<code>speculation_ledger.tsv promise_ledger.jsonl</code>	
<code>python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace</code>	exit 0; events: 239, plan milestones: 46; inherited warnings only
<code>python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace</code>	exit 0; inherited warnings only

The inherited backlog includes the consumed immutable exception for malformed ledger line 85, legacy noncanonical paths, plan pseudo-milestone gaps, orphan reports/final/* artifacts, missing legacy report/manager artifacts, and known root-layout warnings for legacy root files. The audit characterizes these as nonblocking inherited backlog rather than new repair predicates.

The reporter did not rerun the validator suite. The validator outcomes above are reported from the supplied audit.

Artifact Changes

The supplied audit states that no files were edited, rebuilt, or written during the worker/auditor cycle work, and that no ledger rows were promoted.

For this reporter pass, MANIFEST.md was updated as required for report_cycles_47-49. The update preserved the existing ## Key Files section verbatim, changed the report window from cycles 44-46 to cycles 47-49, updated the promise-ledger event count from 237 to 239, and restated the validated no-op terminal-maintenance posture.

No cycle-specific figures were found under reports/cycles/, and no figures are embedded in this report.

Discussion

Cycles 47-49 are a terminal maintenance interval. Their significance is that the campaign did not drift after closure. No track-local hypothesis was promoted, no master-ledger row was added, no validator failure appeared, and no worker/auditor artifact churn occurred.

The audit guidance remains the controlling future-work rule: continue terminal maintenance unless future input supplies a concrete maintenance defect, reproducibility failure, public-facing clarity/accessibility or broken-link issue, or qualifying scientific evidence with auditable validation and ablation controls. Without that predicate, reopening tracks, changing schema or evidence tables, regenerating continuity-only reports, or promoting ledger rows would add process churn rather than scientific progress.

The original research success criterion remains unmet. PhytoGraph produced substantial infrastructure, validation, falsification, public communication, and handoff artifacts, but it did not produce at least one validated prediction per track. The conservative closure state is therefore preserved.

Open Questions

The remaining questions are maintenance gates rather than active research questions:

- Will a future cycle supply a concrete defect, reproducibility failure, public-facing issue, or qualifying scientific evidence?
- If new evidence is supplied, will it satisfy the established reopening predicates before any track status or master-ledger change?
- Can inherited validator warnings remain documented as historical backlog unless a future validation run reports a new error?
- Can public-facing maintenance remain limited to clarity, accessibility, link repair, provenance hygiene, and reproducibility?

References

No external bibliographic references are cited for cycles 47-49. REFERENCES.md was read for citation continuity, but this report cites only local project artifacts and the supplied audit record.

Appendix: Implementation Details

Source Inventory

Source	Date / cycle	Contents	Timeline role
Supplied audit report	Cycles 47-49	Validation summary, no-op decision, ledger counts, validator outcomes, sub-topic assessment, guidance, and cumulative progress notes	Primary validation source
Cycle 47 sessions	Cycle 47	Session IDs supplied; full transcripts unavailable locally	Source gap
Cycle 48 sessions	Cycle 48	Session IDs supplied; full transcripts unavailable locally	Source gap
Cycle 49 sessions	Cycle 49	Session IDs supplied; full transcripts unavailable locally	Source gap
reports/cycles/report_cycles_44-46.md	Prior reporting window	Previous terminal-maintenance report	Immediate continuity source
reports/final_campaign_handoff_manifest.md	Final handoff	Canonical closure state, track statuses, warning posture, and reopening conditions	Closure-boundary source
prediction_ledger.tsv	Current workspace	Header-only master prediction ledger	Non-promotion evidence
speculation_ledger.tsv	Current workspace	Header-only master speculation ledger	Non-promotion evidence

Source	Date / cycle	Contents	Timeline role
promise_ledger.jsonl	Current workspace	239 append-only events	Ledger-state source
MANIFEST.md	Current workspace	Updated snapshot for future researcher/worker turns	Reporter-maintained artifact

Code Organization

No code was added or modified as part of the worker/auditor cycle work. Existing referenced maintenance code remains:

File	Lines	Purpose
scripts/build_taxonomy_results_site_assets.py	428	Builds the public taxonomy-results review site data, evidence tables, and figure assets from the final six-track status table.
tests/test_taxonomy_results_site_public_text.py	130	Verifies required site files, figure references, route labels, public-language boundary, six status codes, and header-only master ledgers.
reports/reopen/scripts/build_final_free_tier_closure_synthesis.py	197	Builds the final six-track free-tier status table, closure report, and summary figure consumed by the handoff and site artifacts.
tests/test_final_free_tier_closure_synthesis.py	109	Verifies the final six-track status table, root/reopen propagation, figure presence, and header-only master ledgers.

File Counts

File	Count
prediction_ledger.tsv	1 line
speculation_ledger.tsv	1 line
promise_ledger.jsonl	239 lines
MANIFEST.md after reporter update	119 lines
REFERENCES.md	157 lines
reports/cycles/report_cycles_44-46.md	192 lines
reports/final_campaign_handoff_manifest.md	76 lines
reports/taxonomy_results_site_closure_note.md	15 lines

File	Count
reports/taxonomy_results_site_qa.md	40 lines

Test Results

The reporter did not rerun the validator suite. The supplied audit reports:

- `wc -l prediction_ledger.tsv speculation_ledger.tsv promise_ledger.jsonl`: 1, 1, and 239.
- `python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace`: exit 0, events: 239, plan milestones: 46, inherited warnings only.
- `python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace`: exit 0 with inherited warnings only.

The reporter performed local line-count and header checks for the master ledgers and confirmed both master ledgers remain header-only.

Cross-Reference Map

Origin	Consumer	Flow
Supplied cycles 47-49 audit report	This periodic report	Establishes the validated no-op terminal-maintenance decision.
reports/cycles/report_cycles_44-46.md	This periodic report	Provides immediate prior terminal-maintenance context.
reports/final_campaign_handoff_manifest.md	Future maintainer turns	Preserves canonical closure artifacts, final track outcomes, warning posture, and reopening condition.
prediction_ledger.tsv and speculation_ledger.tsv	Closure records and future cycles	Header-only state confirms no master prediction or speculation promotion.
promise_check and org_check audit results	Audit decision	Confirm validator success with inherited nonblocking warnings only.
MANIFEST.md	Future researcher/worker turns	Updated to summarize cycles 47-49 while preserving ## Key Files verbatim.

Record Gaps

Full native transcripts for the supplied researcher, worker, and auditor session IDs were not available because session-search/session-fetch tools were not available in this environment.

No pre-existing local reports/cycles/report_cycles_47-49.* artifact was found before this final output. That is expected under the reporter protocol: the orchestrator writes the report markdown and renders the PDF from this output block.

Coherence Review

One coherence pass was completed. The report defines the terminal-maintenance posture, distinguishes no-op validation from scientific progress, states the master-ledger non-promotion rule before applying it, reports inherited warnings as inherited rather than new defects, and marks unavailable session transcripts as a record gap.